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Philosophy and Literature 22.1 (1998) 1-30

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Jane, Meet Charles: Literature, Evolution, and Human Nature

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I

In the social sciences since the 1900s and the humanities since the 1960s, the world and the mind have increasingly been seen as socially, culturally, or linguistically constructed. Culture, not biology, shapes what we are; language, not the world, determines what we think. If we are what culture, convention, discourse, or ideology make of us, then there is no such thing as a universal human nature, and to believe in such a thing would be to commit the naive error, or the reactionary crime, of "essentialism."

When the sociologist Emile Durkheim at the end of last century and the anthropologist Franz Boas at the start of this one tried to sever the study of humanity from biology, they had good intellectual and social reasons for doing so in a world where the muddled and heartless doctrines of Social Darwinism and eugenics held popular and even "scientific" sway. ¹ But there was no such excuse when literature departments around the world became mesmerized by the way Roland Barthes, Jacques Derrida, Michel Foucault, and Jacques Lacan drew on the limited linguistics of Ferdinand de Saussure to concur that "Man does not exist prior to language, either as a species or as an individual." ² The new gospel of Theory, propelled by the messianic self-assurance of its prophets, spread through the humanities and the social sciences. What their converts failed to realize was that to follow parochial Paris intellectual fashion meant not only to exclude the world outside language but to ignore some of the major intellectual developments of our time in the understanding of human nature.

Over the last few years the Theory wave has started to break under **[End Page 1]** the inherent weakness of its arguments and the strength of the welling counter-evidence.³ It has also simply exhausted itself by repetition. How many times do you need to "demonstrate" that what may look like biology is really culture in essay after essay about the social construction of the body?⁴

Those reluctant to read outside Theory's approved reading lists may not be aware of it, but evidence has been accumulating for more than thirty years, and with steadily mounting momentum, that not only is it not the case that biology is a product of culture but that culture is a product and a part of biology, and that it is impossible to explain cultural difference without appreciating the complex architecture of the human mind, of a "human nature [that] is everywhere the same."⁵

The evidence comes from many fields, at first working separately, but now genuinely converging in a way very different from the professed interdisciplinarity of much Theory:⁶ from evolutionary theory,⁷ ethology,⁸ linguistics,⁹ artificial intelligence,¹⁰ neurophysiology,¹¹ evolutionary anthropology,¹² analytic philosophy,¹³ evolutionary epistemology,¹⁴ and many branches of psychology.¹⁵ Among the key factors that led to the emergence of evolutionary psychology as a discipline were ethologists' recognition that behavior is a biological adaptation; neurophysiologists' increasing ability to localize particular brain functions; perceptual psychologists' tracing of the diverse and convoluted tasks of human and other perceptual systems; artificial intelligence's discovery of the complexity of many computational tasks we and other animals perform naturally; developmental psychologists' refining of infant attention experiments that allow them to study the understanding prelinguistic infants have of their world; evolutionary theorists' clarification of the constraints on the evolution of reciprocal altruism; and linguists' discovery of the extent to which language is preprogrammed into the human mind.

The evidence converging from these many sources indicates that there is a universal human nature, reflecting the complex organization of the mind, and that the local differences of culture cannot be explained without understanding our common mental makeup.¹⁶ What seems transparent and natural to us in the way we respond to the world should not be taken as givens. Our senses, emotions, and thoughts are intricately shaped by evolution and in turn shape our world for us. The human mind is not simply a general-purpose mechanism, a formless clay until stamped by culture. Evolution has fashioned the mind in highly specific modes, not simply in the ways we see and hear, but even in terms of the ways we feel, think and interact. **[End Page 2]**

Before explaining this in more detail, let me try to remove some misunderstandings. First, those who argue that culture alone conditions thought often suppose that the only alternative is naive empiricism, the assumption that the world offers itself directly to the mind. That is certainly not the case: the mental modules evolution has assembled are anything but direct or transparent, although they select and transform data with an efficiency and aptness that

make us think we respond to the world "just as it is." ¹⁷

Second, to stress the role of a common human nature does not mean to deny enormous cultural differences. But these differences are made possible by the evolution of the mind. Without the complex shared architecture of the mind, culture would not even be possible; because of that shared design, there are many universals across cultures.

Third, to argue that biology provides a base for human life does not mean that it must impose a model for human morality. There is no reason why an origin must predetermine a particular end, or why *is* should lead to *ought*; biology in any case offers a vast array of different models; and culture is itself a part of biology and has the power to generate a cascade of new possibilities.

Certainly, a modern evolutionary view of humanity should not be confused with the old doctrine of Social Darwinism, of nature red in tooth and claw as a model for society, in which untrammelled competition "naturally" leads to the elimination of the unfit. Nor can it support racism. "On the contrary," Ellen Dissanayake points out, "it insists on the reverse. An evolutionary perspective does not and should not attempt to account for differences among members of a species, but rather should elucidate in what ways one species (e.g., humans) is different from others." ¹⁸ As John Tooby and Leda Cosmides argue, adaptionist approaches, by showing how much all human minds have in common, "offer the explanation for why the psychic unity of humankind is genuine and not just an ideological fiction" and demonstrate why differences between ethnic groups must be peripheral and minor ("Foundations," p. 79).

Nor does an evolutionary view of human nature threaten freedom and social change. In fact it offers a reason to resist the moulding of our minds by those who think they know best for us: the cultural constructionist's view of the mind as a blank slate "is a dictator's dream" (Pinker, p. 427); "if we were a blank slate . . . then it would not matter what people did to us, for we could learn to accept it" (Dissanayake, p. 28). An evolutionary view allows for informed social change. Seeing the world as discourse, as text, has led to the comforting academic idea that **[End Page 3]** it can be transformed just by the act of rereading. But no, there is a world outside language, and that world does call out for social change. As Dissanayake notes, "When we see patterns of behavior that were adaptive in the original environment in which they evolved, and realize that they are maladaptive in present-day circumstances, we can set about trying consciously to change them (by cultural means)" (p. 32).

Fourth, that our minds are a product of evolution does not entail that we are "genetically determined," if that is taken to mean that genes control everything without the environment ever affecting them. Biologists rightly insist that phenotypes are not uniquely determined by genotypes but are always inextricably codetermined by the interacting of genes and the environment. As Richard Dawkins notes, our sex drive presumably comes from our genes, yet we all manage to curb our sexual desires "when it is socially necessary to do so": "it is perfectly possible to hold that genes exert a statistical influence on human behavior while at

the same time believing that this influence can be modified, overridden or reversed by other influences." ¹⁹

That our minds reflect evolution's design does not mean that all is nature and not nurture, that all is heredity and not environment. In any sophisticated biological thinking these oppositions have been thoroughly discredited (Brown, pp. 147-49; *ELT*, pp. 270-71): as a species we are "100% biological and 100% cultural." ²⁰ But what it does mean is that culture is not all; culture is part of biology, not an alternative to it; without innate domain-specific mental mechanisms, we could not acquire the culture we do, and without the input of culture, those mechanisms would not be switched on.

Fifth, for these mechanisms in the mind to be innate does not require that we are born with them all already in place. Genes operate to develop us throughout our lives, and the fact that we are born without adult teeth or facial hair or enlarged breasts does not mean we learn them; nor do we learn our first language so much as develop it. Children acquire language within the same age band whether or not it is the cultural pattern for their parents to speak much to them; deaf children begin to use pronouns in signing at exactly the same time as hearing children do, and for the same short period as hearing children they make exactly the same initial confusions about pronouns in their sign language, signing "you" for "me" and vice versa--which they had never done previously, at the stage of prelinguistic pointing. ²¹

Sixth, some confusions about Darwinian evolution. As late as the 1970s, someone of Piaget's standing could fail to realize that evolution **[End Page 4]** did not work in a Lamarckian way (Piattelli-Palmarini, pp. 370-71), so let me spell it out: it is impossible for any change in the course of one's lifetime (short of genetic damage through radiation or chemical poisoning) to be passed on to one's descendants; behavioral adaptations in a single lifetime cannot affect genes in the next generation. There is no development towards perfection, nor even any design for optimality, but merely selection over time towards better design for particular purposes in a particular environment.

Seventh, evolution is not a matter of Just-So stories, of post-hoc storytelling with no predictive power or chance of falsification. By generating hypotheses about design features needed for minds to have coped with individual adaptive problems in social exchange, for instance, ethologists have been able "to look for--and discover--previously unknown aspects of the psychology of social exchange in species from chimpanzees, baboons and vervets to vampire bats and hermaphroditic coral-reef fish." ²² Evolutionary principles have also predicted and discovered hitherto unsuspected features in the human mind, such as a cheater detection module ²³ or female superiority over males in certain ways of processing spatial information. ²⁴

Human psychological adaptations evolved during the Pleistocene era as solutions to problems that faced our ancestors then: problems like selecting mates, avoiding sex with kin, recognizing faces and emotions, acquiring language, "'mind reading' (i.e. inferring the

motives, intentions, and knowledge of others based on their situation, history and behavior)," ²⁵ engaging in social exchange, avoiding being bitten by snakes, identifying plant foods, selecting habitats, and much more ("Foundations," p. 110; "Intuition," p. 81). They are not necessarily adapted to our present circumstances (the classic example is our sweet tooth, which encouraged our ancestors to eat fruit at the point of maximum nutritiousness but has become maladaptive now that we can manufacture our own sweets) and it seems unlikely that there has been enough time for any further evolution of mental organization to take place in the evolutionarily brief space since the advent of agriculture (we haven't yet had time to lose our craving for sweetness). The past survival value of many of these adaptations may not seem apparent, but when David Premack doubted anyone could explain why elaborate grammar would have been needed in "discussions of mastodon hunting or the like," ²⁶ Steven Pinker and Paul Bloom replied: "it makes a big difference whether a far-off region is reached by taking the trail that is in front of the large tree or the trail that the large tree is in front of. It **[End Page 5]** makes a difference whether that region has animals that you can eat or animals that can eat you. It makes a difference whether it has fruit that is ripe or fruit that was ripe or fruit that will be ripe" (*AM*, pp. 451-93).

These problems to which adaptations are solutions could never have been solved by some general-purpose mechanism in the mind. Often they are problems of which we are unaware or which by definition could not have been solved in a single lifetime. We see movements, shapes and colors automatically, without realizing that our visual processing requires separate motion-, edge- and form-detection mechanisms, separate light-dark and color-detection mechanisms and color-constancy mechanisms. Hamilton's kin selection equation, "that X should help Y whenever $C_x < r_{xy}B_y$," could never have been solved by a general-purpose mechanism in the mind of a hunter-gatherer but was solved and implanted in our emotions by natural selection ("Origins," p. 93). Just as we need not consciously compute Hamilton's equation, we need not consciously strive to maximize our own fitness, since evolution has constructed us so that we feel the urge to act in ways that in average conditions in Pleistocene environments would by and large have contributed to our evolutionary fitness ("Foundations," pp. 111-12).

The result of all these adaptations is a complex mental architecture across the whole species. Just as we are all born with two legs for locomotion and two opposable thumbs for grasping, all our functional psychological adaptations are fixed, not variable, although in non-essentials (height, hair color, shades of personality) we vary within a certain range. ²⁷

As a result of these adaptations there is, despite individual and other variation, a universal human nature, although with of course certain systematic differences between individuals at different stages of life, between male and female (with respect to reproduction and rearing, in terms of aggression and cooperation and spatial sense), and between people in different social situations.

Why has the mind evolved with many particular psychological adaptations or mental modules ²⁸ rather than simply as a general-purpose mechanism to which culture and

circumstance could give detailed form?

The problems even of perceiving the world and moving about in it, let alone of making decisions about what to do in it, have turned out to be vastly greater than anyone realized fifty years ago. Nature has made the solutions so transparent that we often could not even see there were problems. To take a simple example, experiments have shown that we **[End Page 6]** can pick out objects more quickly by color than form;²⁹ yet the spectral reflections off objects alter vastly in the course of a day's changing illuminations. In other words, colors are anything but constant; but because recognizing objects rapidly is of such survival value, and primates, evolving in trees, needed sight more than smell, the primate mind developed to adjust for color constancy immediately, with a sophistication that no artificial system can yet remotely match.

The mind needs to provide specific solutions to specific problems. A general-problem-solving mechanism, as we have come to understand since the advent of computers, simply will not work. "There is no such thing as a 'general problem solver,'" observes Donald Symons, "because there is no such thing as a general problem."³⁰ Just as a computer needs specific software and specific subroutines before it can begin to perform specific tasks, we need specific mental capacities: "The cognitive programs that govern how [we] choose a mate should differ from those that govern how [we] choose [our] dinner" ("Intuition," p. 87). General-purpose learning by association turns out to be far too weak to account for the acquisition of language³¹ or other aspects of culture. As Tooby and Cosmides conclude: "the content-independent psychology that provides the foundation for the Standard Social Science Model is an impossible psychology. It could not have evolved; it requires an incoherent developmental biology; it cannot account for the observed problem-solving abilities of humans or the functional dimension of human behavior; it cannot explain the recurrent patterns and characteristic contents of human mental life and behavior; it has repeatedly been empirically falsified; and it cannot even explain how humans learn their culture or their language."³²

Highly domain-specific cognitive mechanisms have been found in young infants, in "physics, psychology, number, and geometry";³³ and they have been found in adults, cross-culturally: mechanisms for language, for the senses, for different kinds of memory (for faces, things, places), different kinds of emotion (sexual jealousy, parental attachment), different modes of understanding the physical world or other minds or social situations. One example will suggest how specific, unexpected, and high-level they can be.

Considerations from game theory and evolutionary theory showed in the 1960s and 1970s that cooperative social exchange could not evolve as an evolutionarily stable strategy in any species unless individuals had some way of detecting cheaters and withholding future cooperation in retaliation for cheating. **[End Page 7]**

Now a common test of purely logical reasoning, the Wason selection test (1966), normally difficult for most people to solve, turns out to be very easy to pass when it takes the form of

a problem involving cheating in social exchange. In the Wason test, the subject is asked whichever of four cards must be turned over to test a proposition. For instance, if the proposition is "If a person is a Leo, then that person is brave," and there are four cards with a star sign on one side and a rating of courage on the other, the sides turned up being "Leo," "brave," "Aries" and "coward," then most people realize they have to turn over the "Leo" card, many think they also have to turn over the "brave" card, and considerably fewer realize that in fact they need to turn over both the "Leo" and the "coward" cards and no more. If we had special circuits for logical reasoning, we would see that *If P then Q* needs to be tested against *P and not-Q*, but usually "fewer than 10% of subjects spontaneously realize this" ("Intuition," p. 90).

When the form of the question is altered so that the content requires subjects to look for cheaters in social exchange, the performance jumps dramatically to 70% to 90%, the highest ever recorded for the test, even when the situation described is culturally unfamiliar and deliberately bizarre (for instance: "If a man eats cassava root, then he must have a tattoo on his face"). That performances remained high with these unfamiliar scenarios if they were set up as social contracts (but not if they were presented as non-contractual situations based on the same bizarre condition) showed that it was not the familiarity of the content that boosted performance. Leda Cosmides then "tested unfamiliar social contracts that were *switched*" so that the correct social-contractual result was now the wrong logical response. 71% still answered with "the 'look for cheaters' response, *not-P and Q*, . . . even though this response is illogical according to the propositional calculus." ³⁴

People who cannot usually detect violations of "if-then" rules, then, can do so easily when the violation involves cheating in social exchange ("Intuition," p. 91). No one had thought, until evolutionary theory showed the constraints on the evolution of social exchange, to expect a cheater detection module in the mind, and colleagues thought Cosmides crazy even to look. ³⁵ She and others have since found other evidence of human content-dependent reasoning in other domains (aggressive threats, protection from hazards). ³⁶

Cosmides and Tooby show the likelihood of a social grammar innate in the mind in the way Chomsky has shown a linguistic grammar to be. They argue that the Standard Social Science Model "would have to [End Page 8] predict that whatever social exchange is found to exist, it would have to be taught or communicated from the ground up"; but this, they show, is as much at odds with the evidence as the idea that the complexities of language must be taught to infants. "We suggest that social exchange is learned without explicit enumeration of underlying assumptions. The surface forms are provided by specific cultures, but they are interpreted by content-specific information-processing devices containing implicit assumptions and procedures that evolved for exactly this purpose." ³⁷ Just as Chomskyan linguistics has shown that there are sentences that reflect syntactic rules logically inferable from common constructions but that are in fact violations of Universal Grammar and hence never used, even by children, so Cosmides and Tooby show that there are eccentric sentences that "involve no *logical* contradictions . . . [but] seem to violate a grammar of social reasoning--in much the same way that 'Alice might slowly' violates the grammar of English but 'Alice might come' does not" ("Intuition," p. 95). Just as the Universal

Grammar is complex and non-transparent, but appears to be a basis for our all being able to acquire language at the same age, whether or not we are taught, so there seems to be a Universal Grammar "of social reasoning" (p. 100). Without this and other domain-specific "universal reasoning instincts, the acquisition of one's 'culture' would be literally impossible, because one wouldn't be able to infer which representations, out of the infinite universe of possibilities, existed in the minds of other members of the culture" (p. 100).

If much of what we have assumed to be the unitary faculty of "reason" breaks down into local reasoning instincts, the faculty of "emotion" similarly subdivides in evolutionary terms into an array of decision-making impulses, each of which makes something that in the evolutionary past was on average advantageous for us seem desirable or, if disadvantageous, repellent or distasteful: things as diverse as food tastes, landscape preferences, fear of the dark or of snakes, sexual and parental feelings, feelings about status and belonging. ³⁸

The evidence suggests that the mind has evolved as a network of domain-specific modules, "a face recognition module, a spatial relations module, a rigid object mechanics module, . . . a social-inference module, a sexual-attraction module, a semantic-inference module, . . . and so on" ("Foundations," p. 113), and that because we all share this complex mental architecture, we can learn whatever culture we happen to be born into. The minds of human infants are designed to take the inputs of any human culture. Particular switches will be set, particular **[End Page 9]** ranges of options will come into play, according to different cultural input and resulting in different output. From a universal innate psychology emerge the diverse mental settings and behavior of the world's cultures.

In the social constructionist model of culture, cultures themselves, hypostasized as entities with aims and interests and activities of their own, stamp themselves on the putty of the mind. In the evolutionary model, an innate psychology, the product of natural selection, makes it possible for individuals to acquire their particular cultures, according to the situation into which they are born and their perception of their own interests. As Martin Daly and Margo Wilson point out, the constructionist model "cannot be correct, insofar as it elevates the so-called system's objectives above those of its actors and ignores the fact that [the] members [of a social group] are agents with only partially congruent interests. A quarter of a century of criticism of greater-goodism in biology has clarified why individual organisms are the appropriate level in the hierarchy of life at which to impute integrated agendas, and why the analogizing of groups to self-interested individuals typically fails." ³⁹ Foucault and Greenblatt, take note.

Dan Sperber's evolutionary model for culture suggests that to explain culture

is to explain why and how some ideas happen to be contagious. This calls for the development of a true *epidemiology of representations*. . . . All epidemiological models . . . have in common the fact that they explain population-scale macro-phenomena, such as epidemics, as the cumulative effect of micro-processes that bring about individual events, such as catching

a disease. In this, epidemiological models contrast starkly with "holistic" explanations, in which macro-phenomena are explained in terms of other macro-phenomena--for instance religion in terms of economic structure (or conversely). (pp. 1-2)

Since the social constructionist model requires minds without form and populations of individuals without individual interests to learn through passive imitation or absorption what is imposed by a "culture," it cannot in fact explain the origin, transmission or growth of the Culture on which it relies so heavily. ⁴⁰ In evolutionary anthropology by contrast ideas catch on because evolution has built minds they can catch on to. **[End Page 10]**

II

What are the implications of an evolutionary understanding of human nature for literature? This will require much hard thinking; even asking the right questions will be difficult.

What evolution should not do is merely provide a loose metaphor for cultural change, even after Richard Dawkins's suggestion of "memes" as a cultural equivalent of "genes." ⁴¹ Culture does not operate, as evolution does, by means of impersonal selective advantages incorporated into design over thousands of generations; it involves transformation as much as transmission at each step; and it makes possible deliberate design. ⁴²

Against the "Theory" paradigm dominant of late in film criticism, David Bordwell and Noël Carroll have advanced a cognitive approach to film that comes close to the naturalism of an evolutionary perspective. They argue against merely replacing one Grand Theory to which everything needs to be referred with another. Instead, they suggest tackling particular domains (horror, humor, the rationale for shot-reverse shot editing, the role of film music, etc.) with particular functional explanations undecided in advance by some grand theory; they argue against piling up examples from texts or films as if these "proved" the theory, especially when standards of evidence have become as slack as they have in criticism; they counsel against merely taking on another new theory as an easy heuristic tool to offer a new wave of "readings" more or less preordained by their point of origin. ⁴³ All their warnings are well taken.

How can an evolutionary perspective avoid these dangers, and what can it offer literary studies? It can allow us to see human experience, which is after all the subject of literature, in the widest available context, in time--in terms of what brought human nature into being--and in scale, in an understanding of our world that extends from physics to chemistry to biology to psychology to culture. It can allow ways of understanding human minds, as producers and consumers and subjects of literature, that are neither parochial nor deceived by what seems obvious, automatic and "natural." It can help free us from the confused and untenable idea that reality is a linguistic or social construct, while acknowledging that the way we see the world is one that has coevolved with our needs. As Joseph Carroll argues, it can remove the totalizing force of culture in recent theory, and restore the place of

individuals (authors, characters, readers) pursuing their own **[End Page 11]** interests in their environment, an environment that in our case happens to be also richly social and complexly cooperative and competitive. By restating the case for human universalism, it can invite us to look for universals as well as differences in literature: as Dissanayake remarks, it is "especially well suited for multicultural and other approaches to 'non-elitist' arts." ⁴⁴

Literary criticism that starts from an understanding of the evolved human mind should raise many new questions about things we have taken for granted. How do evolutionary, cognitive and developmental psychology suggest we should understand the process of literary composition and comprehension and response? Why do we receive pleasure from literature (why is it *pleasure* that we receive from literature)? Why does it stir the gamut of our emotions? What capacities in us does it activate, and why?

Of course evolution has not selected us as readers of literature, but in all present societies, including hunter-gatherer ones, people derive pleasure from well-turned phrases and stories produced by gifted phrasemakers and storytellers. Out of what behaviors in the past, with what advantages in the Pleistocene environment, could literature have evolved?

Play "has been observed in most, if not all mammals, which have been studied in detail," ⁴⁵ and presumably seems pleasurable because it offers essential practice in real-world skills without real-world risks, whether the practice be in sensorimotor or cognitive skills. "Even animals deprived of the opportunity to play when young," notes Dissanayake, "do not develop normal adult social behavior" (*What Is Art For?*, p. 78) Play helps build not only competence but also flexibility: Michael J. Ellis comments that "The very animals that play most are those animals which exhibit variable and adaptive behavior as adults and tend to inhabit environmental niches where conditions are not fixed." ⁴⁶

Since language is an essential part of being human, we love to play with words. The play starts as parents everywhere slow down their speech, raise its pitch and exaggerate its rhythm when talking to their children, ⁴⁷ and continues into playground jingles and adult odes. Languages the world over have developed verse systems based on lines that take around three seconds to utter, the time psychologists have measured as the unit of the conscious present. ⁴⁸ Meter, rhyme and other forms of repetition stimulate a pleasure in detecting pattern that is of advantage in coping with any aspect of our environment. ⁴⁹ **[End Page 12]**

Since narrative is essential to our sharing and explaining actual experience and simulating potential future experiences, ⁵⁰ we love to play with stories. Pinker brilliantly shows the adaptive advantages of parsing the world into objects and events, and of equipping our language with nouns and verbs to match (*Language Instinct*, pp. 154-56). Even in infancy we soon parse the world much further: we develop sophisticated ways of distinguishing between inanimate and animate objects ⁵¹ and develop a theory of agency. ⁵² We tend to "'see' goal-directed action . . . on the slightest provocation," ⁵³ so that in our stories

purposiveness becomes the default indicator of relevance. Between the ages of two and five, whether in Europe, America, and Asia or in a hunter-gatherer culture in Africa, we develop a theory of mind that allows us to pretend and understand the pretense of others and to understand belief and desire in others ("Origins," p. 102). It has been argued that the advantages in being able to deceive others, and then in turn to detect the subtle signs of someone else's deception, seem likely to have led to a "cognitive arms race" in human evolution, and perhaps even to account for the explosive development of the human frontal lobes.⁵⁴ In stories, we enjoy exercising our ability to "read" the positions of others, to see the power of deception and disguise, to appreciate the dangers of mistaken constructions of a situation or of someone else's state of mind. Tales within tales in traditional stories, and fictional representations of thought in more modern ones, allow us to exercise what Theory of Mind psychologists call the capacity for metarepresentation⁵⁵ that we need in everyday life as we try to react to what we think another person thinks we or someone else may be thinking. Disguise, deception, dramatic irony, tales within tales, and glimpses of characters' thoughts are not merely timelessly popular narrative devices: their presence in stories is, and probably long has been, a way of training us for the complexities of social life.

An evolutionary perspective invites us to see how literature arises out of deep-rooted human needs and capacities, how it works less through the assemblage of arbitrary codes that structuralism saw or the inevitable ideological shaping that poststructuralism sees, than through widely-shared human interpretive competences, linguistic, social, and narrative, through the ordinary, evolved processes of the mind for dealing with everyday reality.

This does not mean that literature offers direct access to reality or even to fiction, but that stories can function with such efficiency that they seem transparent and natural, just as the visual system seems **[End Page 13]** simply to register the actual colors of things. Nor does it mean that even the text simply imprints itself on the mind intact: recent cognitive work suggests strongly that "comprehension is not expression in reverse" and "recall is not storage in reverse." Communication and memory "transform information" (Sperber, p. 31).

Literary understanding arises out of evolved competences we use all the time: perceptual, linguistic, orientational, causal, emotional, social, moral, competences for the identification and recollection and assessment of others, for the reading of emotions, for keeping track (as Frans de Waal writes of the chimpanzees he observes) "of who has done what for whom."⁵⁶ If literature draws on long-established species-wide psychological adaptations, it will surely be more accessible and less culture-bound and more determinate in its meanings than much current criticism suggests (*ELT*, p. 159). This does not mean we each see every work of literature exactly or in exactly the same way; cognitive psychology suggests that representations are consistently transformed in the process of transmission "in the direction of contents that require less mental effort and provide greater cognitive effects. This tendency to optimize the effect-effort ratio . . . drives the progressive transformations of representations within a given society towards contents that are relevant in the context of one another" (Sperber, p. 53). We make mental summaries as we read, a nested series of summaries; the mind does not so much remember subordinate levels of detail as illustrate them in ways that seem plausible from the level above,⁵⁷ and in ways likely to be skewed in

the direction of the familiar; but this natural process need not imply a culture-wide ideological distortion.

Approaching literature through evolution allows us to understand the subject matter of literature in terms of the interests evolution has given us, as well as the way these are further reframed by each culture and each author. Narrative selection must reflect natural selection, human interests and inferences, human ways of dividing up the world (see *ELT*). At the same time, through the course of cultural history and the force of individual genius, literature has extended the range that evolution made natural. At first, before mythmaking and science had begun to separate, we offered explanations for what lies behind the visible, in terms of the animate agency that evolution conditioned us to see as fundamental to causality. ⁵⁸ Then, as science extended our immediate and natural human range to other senses, energies, and scales, literature too evolved, towards effects of light or color or landscape beyond the formulaic, characterization beyond the schematic, **[End Page 14]** dialogue beyond the rhetorical, plot beyond raw impact and cooked contrivance; towards complexities of expression achievable only through writing, nuances of mentation too private and capricious for normal disclosure, subtleties of interrelationship between perception and thought and language, sensitivities of action and reaction.

An evolutionary approach to literature can ask again old questions, recently deemed outmoded, that can now be rephrased in new terms. How do the culturally specific and the humanly universal combine within specific works or traditions? (See *ELT*, esp. pp. 170-71.) How do the alleged universality of the mind and the specificity of cultures intersect in cross-cultural reading? How do traditions spread, for writers and readers, through a process of cultural transmission that allows for the role of the individual mind in accepting, transforming and retransmitting ideas? (See Sperber; Boyer, *Naturalness*.) What in the spread of particular modes and genres and techniques can be attributed to their centrality to evolved human needs? ⁵⁹

If evolutionary science is at last starting to account for and set in context the human experience that has always been at the heart of literature, what role does that leave literature itself? I would answer: while science searches for the laws by which things combine, literature celebrates and consists of irreplaceable particular combinations; while science aims at coherent and convergent explanation, literature results from and invites divergence, an explosion of perspectives and purposes in authors, characters, and readers.

Just as a scientific and evolutionary approach to the mind moves towards universals--universals that link one human mind to another, and then human to animal, animate to inanimate, and ultimately the physical world to the abstract space of logical and mathematical possibility--an artistic or literary approach moves towards specifics: *this* reader encountering *this* writer inventing *this* character in *this* scene.

But if writers seize on the specific, or on the general via the specific, they can do so only through what we share. Idiosyncrasy is not art: in order to attract our interest and allow our understanding, writers have to work with what they and we can understand of human nature.

The broader their sense of human nature, and the more specific their characters and situations, the wider their impact. **[End Page 15]**

III

Let me take as an example a writer who is notorious for the narrowness of her compass--her little bit of ivory, ⁶⁰ her three or four families in a country village ⁶¹ --and who nevertheless currently enjoys an exposure to audiences around the world rivaled only by Shakespeare.

Jane Austen has a restricted range; and although she loves to talk in universals, in abstractions, they are universals that seem conditioned by the very local culture of her time, her place, and her class. How then can she have such wide appeal?

She is less specific--she uses less descriptive detail--than other novelists. She seems singularly uninterested in the world of nature, even in the appearance of her characters, let alone their wider social environment, let alone the nonhuman world around that. It is hard to imagine anyone who shuts the door more firmly on biology and the physical world, hard to imagine anyone further from Darwin. Yet what look like the sheltered mores of a small section of Regency society can be explained in part by principles extending deep into the evolutionary past. Let us look at the Austen novel with the least lively of her heroines, the one least at home in the physical world: Fanny in *Mansfield Park* (1814). ⁶²

All Austen's novels are love stories. Their power depends on the universal and central human problem--a problem we share with most of the animal kingdom--of choosing and winning the right sexual partner. ⁶³ She makes that one choice more central to her work and more urgent than any other major novelist. She hints at the right choice for her heroine soon enough, but she can withhold its realization from one of the partners to within a few pages of the end. We look forward to the right resolution with all the interest we take in so crucial an outcome, but we can never be quite sure we will reach it. Austen uses all her storyteller's arts to create suspense, to block, to sidetrack, to offer alternative resolutions that come dangerously close to realization. She multiplies options and obstacles by multiplying the number of marriageable young men and women around. As in our evolutionary past and our immediate present, sexual choice takes place in a crowded competitive pool.

Austen focuses overwhelmingly on female choice, whose importance Darwin suggested is a driving force behind sexual selection. ⁶⁴ His argument was resisted until the 1970s and 1980s by male scientists who could not believe females could have such power in the mating **[End Page 16]** process. ⁶⁵ A key change came in 1972 when the evolutionary theorist Robert Trivers proposed a definition of "parental investment" as "*any investment by the parent in an individual offspring that increases the offspring's chances of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring*" ⁶⁶ and noted that in all species where the female's investment in the production of children is greater than the males (and that is in almost all animal species), the females select from the

available males.

In humans and other species where the rearing of the young requires a high investment from the male as well as the female--where there is a high Male Personal Investment or MPI--females choose males as partners on the basis of their ability to support the offspring.

⁶⁷ In our own case, recent cross-cultural studies have shown that when it comes to choosing partners, women across the world place a very great stress, much more than men do, on a candidate's prospective resources (Buss, p. 25). In *Mansfield Park* the first character to get engaged, the one with first choice, is Maria Bertram. Whom does she agree to marry? Mr. Rushworth, who just happens to be the richest young man around.

Whether or not there is a high or low MPI, males tend to do the chasing but females do the choosing. And in humans, as Donald Symons observes in *The Evolution of Human Sexuality*, "Heterosexual relations are structured to a substantial degree by the nature and interests of the human female." ⁶⁸

Austen may seem a long way from biology, but she repeatedly focuses on something as elemental to our species as female choice: Elizabeth Bennett chooses Darcy over the unpromising Collins and the initially alluring Wickham; Emma stands firm against Mr. Elton and entertains thoughts of Frank Churchill before realizing under what she thinks is the threat of competition from Harriet Smith that she wants Knightley. Despite her normal meekness, Fanny Price, although strongly pressured by her suitor himself and her own most powerful relatives to accept Henry Crawford, insists on declining him, since she can already see someone she could trust far more as a partner and a father to her children; and she is proven right to do so.

Fanny's choice is the key one in *Mansfield Park*, but Austen doesn't just follow this one line, in the wish-fulfilment way of romance. She looks around.

"In a high-MPI species such as ours, where a female's ideal is to *monopolize* her dream mate--steer his social and material resources toward her offspring--competition with other females is inevitable" **[End Page 17]** (Wright, p. 63). This forms much of the early drama of the novel, in the tension between Julia and Maria as they compete for the favors of Henry Crawford. Maria's opting for the first rule of thumb, good resources, has proved a mistake: Rushworth is not worth rushing into marriage with, since although he has money in the bank he hasn't a thought in his head. When another young man with more than ample resources turns up, and shows he also has brains and an alluringly seductive manner, Maria is ready to think again, to compete even with her own unattached sister. Women the world over prize intelligence (and hence the ability to solve problems) and verbal skills (and hence the ability to persuade others) in prospective partners: ⁶⁹ not that they consciously select men on these terms in order to maximize their reproductive fitness: nature has instead incorporated direct positive cues into the emotions. Indeed, Geoffrey Miller speculates that the rapid evolution of human intelligence may be a product of sexual selection by both male and female: "The neocortex is largely a courtship device to attract and retain sexual mates: its specific evolutionary function is to stimulate and entertain other people, and to assess the

stimulation attempts of others" (cited in Ridley, p. 327).

Fanny looks on aghast at Maria and Julia vying for Henry's favors, but she stays out of the competition herself. Although unattached, she seems too young, too reticent, too lacking in the vitality that for men is a natural sign of a woman's child-rearing potential ⁷⁰ for Henry Crawford even to look at her.

But she has her own interests, her own desires: she loves Edmund, who alone has shown a real care for her since her arrival at Mansfield Park. ⁷¹ But like Maria and Julia competing over Crawford, she does not have the field to herself. Mary Crawford seems to have every advantage over Fanny: age (around the world, the mean difference between men in their early twenties and the ages of the women they desire most is two and a half years [Buss, p. 51]: Edmund happens to be 24, Mary Crawford 21, Fanny only 18 when the main action starts); health; vivacity; wit; status. Much of the poignancy of the novel comes from Fanny's being in love with a man she feels herself not even in a position to compete for: not only can she not vie with Mary's obvious advantages, she does not even dream she might try. Humans, like other apes, tend to be hierarchical, and Mrs. Norris's drubbing into Fanny her social position as the poor niece has reinforced her natural withdrawnness. She can only wait and watch as Edmund falls under Mary's spell and fails to see the signs of the difference in values between them that she can see. **[End Page 18]**

But then the drama shifts. Henry Crawford leaves Mansfield and does not hurry back. He has not been serious about either Maria or Julia.

In most sexually reproducing species males have two main strategies: to fertilize as many females as possible; or to stay with one and raise their joint offspring together. In a high MPI species, such as humans (and many species of birds), the latter will tend to be the preferred option, although "the 'optimal male course,' as Trivers noted, is a 'mixed strategy.' Even if long-term investment is [a male's] main aim, seduction and abandonment can make genetic sense, provided it doesn't take too much, in time and other resources, from the offspring in which the male does invest" (Wright, p. 61).

Henry Crawford now returns to Mansfield after Maria, indignant at his suddenly dropping her, opts out of spite to go through with her marriage to Rushworth. Julia accompanies the newlyweds on their travels, so that Fanny remains the only eligible young woman left at Mansfield. Henry decides to amuse himself by trying to make her fall in love with him.

His frank desire to have women love him--Julia, Maria, Fanny, and reputedly many more in his recent past in London--reflects one side of the male sex drive. He does not bed these women, but simply dallies with them; his sexual drive is contained within the norms of his society--although only for the time being, as it will turn out. Someone who was a thoroughgoing rake, of course, would not be harbored for long in the strict world of Mansfield, and would never be pressed on Fanny as a prospective partner.

Henry may not be an out-and-out reprobate, but in his relations with women he is a

practiced cheat: he aims for the pleasures of feeling loved without the cost. As we have seen, the ability to cheat and to detect cheating appears to have exerted a major pressure on the evolution of human intelligence. Now Trivers suggests that if indeed "deceit is fundamental to animal communication, then there must be strong selection to spot deception, and this ought, in turn, to select for a degree of self-deception, rendering some facts and motives unconscious so as not to betray--by the subtle signs of self-knowledge--the deception being practiced" (cited in Wright, p. 264).

At first Henry plans only to make Fanny fall for him because he finds the challenge amusing. She has spotted his deception of Maria and Julia, and he is a sensitive enough reader of others to detect her disapproval.

But he begins to feel himself genuinely in love with her. Men may be **[End Page 19]** inclined by evolution to scatter their seed widely, but they have also been shaped to be strongly attracted to, to fall in love with, a woman who is likely to prove reliably faithful (Buss, p. 70). Henry finds himself powerfully stirred by Fanny's very firmness, her rock-like integrity. Aware that women appreciate kindness and understanding, ⁷² and knowing that the only anxiety the selfless Fanny will admit to is her favorite brother's stalled career prospects in the navy, he shows her his solicitude and the advantages of his high status when he wangles a promotion for William. He proposes, she resists, he insists, and all the pressure of Mansfield is brought to bear on her. Meekly complicit although she has always been until now, she still resists. Studies show and evolution has registered that the single best indicator of future infidelity is past permissiveness (Buss, p. 69), and although Henry has dabbled only in emotional, not sexual, indiscriminateness, Fanny has noted it well.

By this point Henry genuinely believes himself a changed man. He persists in his wooing, but he is deceived in himself. In London, meeting a Maria still angry with him, he cannot resist trying to rekindle her passion. He succeeds all too well, and adultery ensues. As Matt Ridley observes in *Sex and the Evolution of Human Nature*, "we are designed for a system of monogamy plagued by adultery." ⁷³

Fanny has been able to read Henry in a sense even better than he could himself. If the gene pool tends to favor genes for effective deception by males, and even effective self-deception, natural selection nevertheless tends "to favor females who become good at seeing through such deception" (SG, p. 155). And it is Fanny whom life seems to favor by the end of *Mansfield Park*.

As females become better at seeing through male deception, ⁷⁴ "genes for honest, good fatherhood will come to prevail in the gene pool" (SG, p. 155). Fanny has been attracted to Edmund's sensitivity and scrupulousness from the first, and despite all Henry's attentions, she never wavers. Males too of course need to test for the prospective fidelity of their life-partners (Buss, p. 70). As his sisters were charmed by Henry Crawford's liveliness, Edmund has been charmed by Mary's (and critics by them both), but when Mary's indulgence toward her brother's adultery suggests to Edmund that she may be unreliable as a partner, he at last stops trying to persuade himself that despite their other differences she is the woman he

must ask to marry.

He expects he will never find another: "Scarcely had he done regretting Mary Crawford, and observing to Fanny how impossible it **[End Page 20]** was that he should ever meet with such another woman, before it began to strike him whether a very different kind of woman might not do just as well--or a great deal better; whether Fanny herself were not growing as dear, as important to him in all her smiles, and all her ways, as Mary Crawford had ever been; and whether it might not be a possible, an hopeful undertaking to persuade her that her warm and sisterly regard for him would be foundation enough for wedded love." ⁷⁵

Another biological point. We are a very social species. Other higher primates are already highly social, but because of language, we have an even more finely-attuned response to one another than other primates do, and we constantly have to make sensitive and flexible choices as we interact with others.

Never do we attend to each other with finer tuning than during the highly charged process of sexual selection. The whole situation maximizes the need to reveal and conceal emotions and motives, to read them in others, and to tally one's readings over time; it maximizes the pressures that have evolved to make us such subtle social and moral animals. ⁷⁶

If Austen pays little attention to her characters' physical appearance beyond noting whether they are attractive or not, that is in fact a good approximation of the dynamics of sexual scrutiny, where a provisional rating suffices (impossible, passable, pursuable, unattainable). But she more than compensates by the attention she then pays to the minute details of characters' behavior and speech that she shows can influence or override first impressions. ⁷⁷

Austen is not the only one who attends to these overt details and the intentions they reveal and conceal. Her characters too, unless they are as dim as Mr. Rushworth or as blinkered by self as Mrs. Norris, continually try to read one another's minds. Although Sir Thomas can construe others moderately well, it is the four key lovers who excel. Mary Crawford has a nimble and shrewd alertness to others. Henry is still shrewder, even if with more thoughtlessness too. Edmund has a genuine, mature sensitivity to other minds, and ahead even of him is Fanny, the one who notices everybody but whom everybody was inclined not to notice.

A story and a heroine that lack surface vivacity compensate for it through one poignant and protracted irony: the disparity between Fanny's penetrating insight into everyone else and their failure, even Edmund's failure, to read her great secret, her love for Edmund.

Neglected at first, Fanny triumphs at last. Despite all her initial **[End Page 21]** disadvantages of situation, she lands the best man around, the staunchest partner, through her superior ability to read the minds of others--to see Henry's weaknesses, to see that no one else comes close to matching Edmund's strengths--and through Edmund's ability in turn

to read her superior capacity for reading others.⁷⁸ In this light *Mansfield Park*, like other Austen novels, seems almost an evolutionary romance, in which the cognitive arms race will be won by the socially sensitive. There's more to Austen than conservatism; she's also an evolutionary optimist.

Austen harnesses the dynamism of sexual selection to rivet the attention of readers as she explores the subtle attentiveness to others that is central to her moral imagination. But sexual selection also appears to enrich not only her audience appeal and her subject but even her literary technique.

In Austen the plots, because they are love stories, advance by an intricate integration of action and reaction, observation and reflection. Characters watching each other as carefully as Austen's lovers do need to calibrate their present response to each other with their past reactions, even as the conversation flows on or the situation alters. To render this interplay of external and internal, of mouthspeaking and mindreading, Austen became the first to develop sustained free indirect discourse,⁷⁹ a style that flexibly and efficiently echoes characters' thoughts as they assess each other, or as one person assesses another's response to a third, while maintaining the narrative flow. No one before Austen needed free indirect style as a ready technical resource, because no one before Austen paid such minute attention to the way we monitor ourselves and each other so finely.

Precisely because selecting sexual partners is so crucial and fraught a part of human life, it has always been subjected to complex cultural controls and variations. There are sexual universals in human life, but there is also an astonishing degree of variety from society to society and era to era.⁸⁰ Now we can easily recognize the accuracy with which Austen defines the special conditions of her particular milieu. It may even seem as if she accepts these local conditions as universal. We of course don't: we don't still see marriage in terms of the mores of the genteel countryside of Regency England. But I suspect that Austen too doesn't see the standards of her milieu as universal, but that she knows she can be true to the particulars of her small society, her three or four families in a country village, because she writes about something so essential in human life. She knows just what her focus on the choosing of partners will do for the minds of her characters and her readers. **[End Page 22]**

Why have I offered the example of Austen? This discussion is neither a rounded, let alone a complete, "reading" of *Mansfield Park*, nor a model for future "evolutionary" readings of literature. It stresses Fanny's strength, and women's, and Austen's, but it says nothing about particular literary or historical contexts, and very little about Austen's artistic powers and problems, about her way of individualizing, analyzing, ironizing. I want merely to show how much must be distorted by an argument that presents this or any novel exclusively in terms of the social construction of desire and difference.

I suggest that even in the work of a writer who seems aloof from biology and remote from the humanly universal, much can be interpreted in terms of biologically evolved characteristics of human life, rather than as no more than the product of a particular cultural moment. By seeing Austen's stories in these terms we can explain their power and range,

whereas an explanation in terms only of local cultural conditions can make them seem stifling and airless.

I have tried too to suggest how a thoroughly original mind working from a restricted range of experience can reach through to central, universal, human impulses in order to transform the way stories are told and the way we see each other. Austen draws deep on our whole biological heritage in order to point us towards new stages in the evolution of sensibility. She does not ignore biology; no major writer can; she takes it on, and she subjects it to her own powerful transformation. Accepting the place of biology does not mean rejecting culture: it makes culture possible.

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Notes

1. Durkheim, *Les Règles de la méthode sociologique* (1895; Paris: Flammarion, 1988); Boas, *The Mind of Primitive Man* (1911; New York: Macmillan, 1938). For a detailed discussion of Boas's work as a reaction to eugenics, see Derek Freedman, *The Making and Unmaking of an Anthropological Myth* (1983; Harmondsworth: Penguin, 1996), chaps. 1-4.
2. Roland Barthes, "To Write: An Intransitive Verb?," in *The Languages of Criticism and The Science of Man: The Structuralist Controversy*, ed. Richard Macksey and Eugenio Donato (Baltimore: Johns Hopkins University Press, 1972), pp. 134-56, p. 135.
3. See for instance Noël Carroll, "Prospects for Film Theory: A Personal Assessment," in *Post-Theory: Reconstructing Film Studies*, ed. David Bordwell and Noël Carroll (Madison: University of Wisconsin Press, 1996); and Frederick Crews, "The End of the Poststructuralist Era," in *The Emperor Redressed: Critiquing Critical Theory* (Tuscaloosa: University of Alabama Press, 1995), ed. Dwight Eddins, pp. 45-61. Among scores of other critiques of Theory, some of the most wide-ranging of recent books include Brian Vickers, *Appropriating Shakespeare: Contemporary Critical Quarrels* (New Haven: Yale University Press, 1993); *After Poststructuralism: Interdisciplinarity and Literary Theory* (Evanston: Northwestern University Press, 1993), ed. Nancy Easterlin and Barbara Riebling; Joseph Carroll, *Evolution and Literary Theory* (Columbia: University of Missouri Press, 1995, hereafter *ELT*); and *Beyond Poststructuralism: The Speculations of Theory and the Experience of Reading*, ed. Wendell V. Harris (University Park: Pennsylvania State University Press, 1996).
4. Enough times, it seems, to allow you to convince students that "the pain of childbirth is socially constructed by patriarchy" or that "there was little infant mortality in the past until childbirth was 'medicalized' by men" (cited in Daphne Patai and Noretta Koertge, *Professing Feminism: Cautionary Tales from the Strange World of Women's Studies* [New York: Basic Books, 1994], p. 139).
5. John Tooby and Leda Cosmides, "The Psychological Foundations of Culture" (hereafter

- "Foundations") in *The Adapted Mind: Evolutionary Psychology and the Generation of Culture* (New York: Oxford University Press, 1992), ed. Jerome H. Barkow, Leda Cosmides, and John Tooby (hereafter *AM*), pp. 19-136, p. 38.
6. For the disciplinary convergence in evolutionary psychology, see Leda Cosmides and John Tooby, "Origins of Domain Specificity: The Evolution of Functional Organization" (hereafter "Origins") in *Mapping the Mind: Domain Specificity in Cognition and Culture* (Cambridge: Cambridge University Press, 1994), ed. Lawrence A. Hirschfeld and Susan A. Gelman, pp. 85-116, p. 85. For the lack of genuine interdisciplinarity in Theory, see Easterlin and Riebling, especially Richard Levin, "The New Interdisciplinarity in Literary Criticism," pp. 13-43.
7. In the 1930s Charles Darwin's own ideas were combined with Mendelian genetics to produce the so-called Modern Synthesis, which has been further refined especially since the 1960s in the work of biologists like John Maynard Smith, George Williams, William Hamilton, and Robert Trivers.
8. Deriving especially from the work of Konrad Lorenz and more recently of primatologists such as David Premack and Frans de Waal.
9. Noam Chomsky's work, ever since his *Syntactic Structures* (The Hague: Mouton, 1957), has been crucial; see also Steven Pinker, *The Language Instinct: The New Science of Language and Mind* (1994; Harmondsworth: Penguin, 1995).
10. Especially in the work of Robert Axelrod (see his *The Evolution of Co-operation* [New York: Basic Books, 1984]) and David Marr, *Vision* (San Francisco: Freeman Cooper, 1982).
11. See for instance Daniel C. Dennett, *Consciousness Explained* (Boston: Little, Brown, 1991); Michael S. Gazzaniga, *Nature's Mind: The Biological Roots of Thinking, Emotions, Sexuality, Language and Intelligence* (1992; Harmondsworth: Penguin, 1994); *The Cognitive Neurosciences* (Cambridge: Bradford/MIT, 1994), ed. Gazzaniga; Antonio R. Damasio, *Descartes' Error: Emotion, Reason and the Human Brain* (1994; London: Macmillan, 1996).
12. See Donald E. Brown, *Human Universals* (Philadelphia: Temple University Press, 1991) and the work of Dan Sperber, especially *Explaining Culture: A Naturalistic Approach* (Oxford: Blackwell, 1996), Scott Atran, especially *Cognitive Foundations of Natural History: Towards an Anthropology of Science* (Cambridge-Paris: Cambridge University Press-Maison des Sciences de l'Homme, 1990), and Pascal Boyer, especially *The Naturalness of Religious Ideas: A Cognitive Theory of Religion* (Berkeley: University of California Press, 1994).
13. Especially the work of W. V. O. Quine, in particular *Word and Object* (Cambridge: MIT Press, 1960).

14. Especially Karl Popper, *Objective Knowledge: An Evolutionary Approach* (London: Oxford University Press, 1972).

15. In behavioral psychology, especially the work of John Bowlby; in comparative psychology, that of David Premack; in cognitive psychology, that of Jerry Fodor; in developmental psychology, that of Alan Leslie, Elizabeth Spelke, and Rochel Gelman; in evolutionary psychology, that of Donald Symons, Martin Daly and Margo Wilson, John Tooby and Leda Cosmides; in perceptual psychology, that of Roger Shepard and David Marr.

16. Underlying local variations, "all humans share certain views and assumptions about the nature of the world and human action by virtue of . . . universal reasoning circuits" (Leda Cosmides and John Tooby, "Beyond Intuition and Instinct Blindness: Toward an Evolutionarily Rigorous Cognitive Science" [hereafter "Intuition"] in *Cognition on Cognition*, ed. Jacques Mehler and Susana Franck [Cambridge: Bradford/MIT, 1995], pp. 69-105, pp. 93).

17. "To find someone beautiful, to fall in love, to feel jealous, to experience moral outrage, to fear disease, to reciprocate a favor, to initiate an attack, to deduce a tool's function from its shape--and a myriad other cognitive accomplishments--can seem as simple and automatic and effortless as opening your eyes and seeing. But this apparent simplicity is possible only because there is a vast array of computational machinery supporting and regulating these activities" ("Intuition," p. 99).

18. *What is Art For?* (Seattle: University of Washington Press, 1988), p. 28.

19. Dawkins, *The Selfish Gene* (1976; rev. ed., Oxford: Oxford University Press, 1989; hereafter, *SG*), pp. 332, 331.

20. Daniel G. Freedman, cited in Robert Storey, "'I Am Because My Little Dog Knows Me': Prolegomena To a Theory of Mimesis," in Easterlin and Riebling, pp. 45-70, p. 62.

21. Massimo Piattelli-Palmarini, "Ever Since Language and Learning: Afterthoughts on the Piaget-Chomsky Debate," in Mehler and Franck, pp. 361-93, p. 382.

22. Leda Cosmides and John Tooby, "From Function to Structure: The Role of Evolutionary Biology and Computational Theories in Cognitive Neuroscience" (hereafter "Function"), in Gazzaniga 1994, pp. 1199-1210, p. 1203.

23. Leda Cosmides, "The Logic of Social Exchange: Has Natural Selection Shaped How Humans Reason? Studies with the Wason Selection Task," *Cognition* 31 (1989): 187-276.

24. Irwin Silverman and Marion Eals, "Sex Differences in Spatial Abilities: Evolutionary

Theory and Data," in *AM*, pp. 533-49.

25. Tooby and Cosmides, "Mapping the Evolved Functional Organization of Mind and Brain," in Gazzaniga 1994, pp. 1185-97, p. 1189.

26. "'Gavagai!' or the Future History of the Animal Language Controversy," *Cognition* 19 (1985): 207-96, p. 281.

27. Most variation is individual variation within the same ethnic group (see Pinker, *Language Instinct*, p. 430), now generally explained as a product of the need to resist parasites: see John Tooby and Leda Cosmides, "On the Universality of Human Nature and the Uniqueness of the Individual: The Role of Genetics and Adaptation," *Journal of Personality* 58 (1990): 17-67, pp. 32-35.

28. *Module* is the term favored by David Marr, *Vision*, and by Jerry A. Fodor, *The Modularity of Mind: An Essay on Faculty Psychology* (Cambridge: MIT/Bradford, 1983); Noam Chomsky refers to *mental organs* or *cognitive competences*, in *Reflections on Language* (New York: Random House, 1975); psychologists prefer *domain specificity*, and evolutionary biologists and psychologists talk of *psychological adaptations*.

29. Roger N. Shepard, "The Perceptual Organization of Colors: An Adaptation to Regularities of the Terrestrial World?" in *AM*, pp. 495-532, pp. 509-10.

30. "On the Use and Misuse of Darwinism in the Study of Human Behavior," in *AM*, pp. 137-59, p. 142.

31. "Intuition," p. 98; Pinker, *Language Instinct*, p. 408.

32. "Foundations," p. 114; see Brown, p. 84, for the difficulties met in publishing experimental results that discredited the dogma that anything can be learned by association.

33. Elizabeth Spelke, "Initial Knowledge: Six Suggestions," in Mehler and Franck, pp. 433-47, p. 435.

34. Cosmides and Tooby, "Evolutionary Psychology and the Generation of Culture, Part II. Case Study: A Computational Theory of Social Exchange," *Ethology and Sociobiology* 10 (1989): 51-97, p. 91.

35. Her findings have since been repeatedly confirmed; see especially Gerd Gigerenzer and Klaus Hug, "Domain-specific Reasoning: Social Contracts, Cheating, and Perspective Change," *Cognition* 43 (1992): 127-71.

- [36.](#) See "Function," p. 1203; Tooby and Cosmides, "Think Again," in *Human Nature: A Critical Reader* (New York: Oxford University Press, 1997), ed. Laura Betzig, pp. 292-94, p. 293.
- [37.](#) "Cognitive Adaptations for Social Exchange," in *AM*, pp. 163-228, p. 208.
- [38.](#) Tooby and Cosmides, "The Past Explains the Present: Emotional Adaptations and the Structure of Ancestral Environments," *Ethology and Sociobiology* 11 (1990): 375-424, especially pp. 407-17; Robert Wright, *The Moral Animal: Evolutionary Psychology and Everyday Life* (1994; London: Abacus, 1996), p. 175; Damasio, chaps. 6-8; see also Dissanayake, p. 27.
- [39.](#) "Discriminative Parental Solitude and the Relevance of Evolutionary Models to the Analysis of Motivational Systems," in Gazzaniga 1994, pp. 1269-86, p. 1283; see also Daly and Wilson, "Homicide and Cultural Evolution," *Ethology and Sociobiology* 10 (1989): 99-110, pp. 108-09; "Foundations," p. 47; and *ELT*, p. 151, citing Donald Symons and Donald Brown.
- [40.](#) As Tooby and Cosmides note: "Rejecting the design of individuals as central to the dynamics [of culture] is fatal to these models, because superorganic (that is, population-level) processes are not just 'out there,' external to the individual. Instead, human superindividual interactions depends intimately on the representations and other regulatory elements present in the head of every individual involved" ("Foundations," p. 47). See also Boyer, p. 22.
- [41.](#) *SG*, chap. 11; see also Daniel C. Dennett, "Mememes and the Exploitation of Imagination," *Journal of Aesthetics and Art Criticism* 48 (1990): 127-35.
- [42.](#) Dawkins, *The Extended Phenotype: The Long Reach of the Gene* (Oxford: Oxford University Press, 1982), p. 112, himself notes these differences.
- [43.](#) *Post-Theory*, especially Carroll, "Prospects," pp. 43-44; see also Bordwell, *Making Meaning: Inference and Rhetoric in the Interpretation of Cinema* (Cambridge: Harvard University Press, 1989); Carroll, *Mystifying Movies: Fads and Fallacies in Contemporary Film Theory* (New York: Columbia University Press, 1988) and *Theorizing the Moving Image* (Cambridge: Cambridge University Press, 1996).
- [44.](#) "Darwin Meets Literary Theory," *Philosophy and Literature* 20 (1996): 229-39, p. 235.
- [45.](#) Peter K. Smith, "The Study of Play in Mammals," in *Play in Animals and Humans* (London: Blackwell, 1984), ed. Peter K. Smith, p. 67.
- [46.](#) *Why People Play* (Englewood Cliffs: Prentice-Hall, 1973), p. 41.

[47.](#) Anne Fernald, "Human Maternal Vocalizations to Infants as Biologically Relevant Signals: An Evolutionary Perspective," in *AM*, pp. 391-428.

[48.](#) Frederick Turner and Ernst Pöppel, "The Neural Lyre: Poetic Meter, the Brain, and Time," *Poetry* 142 (1983): 277-309.

[49.](#) John D. Barrow, *The Artful Universe* (1995; Harmondsworth: Penguin, 1997), p. 105.

[50.](#) See *SG*, p. 59, on the advantages of simulation and its likely role in the evolution of consciousness.

[51.](#) Rochel Gelman, Frank Durgin, and Lisa Kaufman, "Distinguishing Between Animates and Inanimates: Not By Motion Alone," in *Causal Cognition: A Multidisciplinary Debate* (Oxford: Clarendon, 1995), ed. Dan Sperber, David Premack, and Ann James Premack, pp. 150-84.

[52.](#) Alan M. Leslie, "ToMM, ToBY, and Agency: Core Architecture and Domain Specificity," in Hirschfeld and Gelman, pp. 119-48, and "A Theory of Agency," in Sperber, Premack, and Premack, pp. 121-41.

[53.](#) David Premack and Ann James Premack, "Moral Belief: Form versus Content," in Hirschfeld and Gelman, pp. 149-68, p. 150.

[54.](#) Robert Trivers, "The Evolution of Reciprocal Altruism," *Quarterly Review of Biology* 46 (1971): 35-57; see also *SG*, pp. 187-88; Richard Dawkins and John Krebs, "Animal Signals: Information or Manipulation?," in *Behavioural Ecology: An Evolutionary Approach* (Oxford: Blackwell Scientific, 1978), ed. John Krebs and Nicholas Davies, pp. 282-309; *AM*, p. 483.

[55.](#) Alan Leslie, "Pretense and Representation: The Origins of 'Theory of Mind,'" *Psychological Review* 94 (1987): 412-26, and "A Theory of Agency."

[56.](#) *Good Natured: The Origins of Right and Wrong in Humans and Other Animals* (Cambridge: Harvard University Press, 1996), p. 24.

[57.](#) Frederic C. Bartlett, *Remembering: A Study in Experimental and Social Psychology* (Cambridge: Cambridge University Press, 1932), suggested the reconstructive nature of memory and introduced the notion of story gists; Ulric Neisser, "What is Ordinary Memory the Memory Of?," in *Remembering Reconsidered: Ecological and Traditional Approaches to the Study of Memory* (Cambridge: Cambridge University Press, 1988), ed. Ulric Neisser and Eugene Winograd, pp. 356-73, discusses the "nesting" of events on pp. 361-67 and introduces the idea of memorial "illustration" on p. 367.

58. David Premack and Ann James Premack stress the "ontogenetic priority of intentional" over mechanical cause, "Afterword," in Sperber, Premack, and Premack, pp. 651-54, p. 652. Boyer points out that while religious beliefs single out the unnatural and therefore attention-getting, they in fact presuppose common-sense understandings of the world and supervene on rather than displace solid-object physics, normal notions of agency, etc.: see Boyer, *Naturalness*, and "Causal Understandings in Cultural Representations: Cognitive Constraints on Inferences from Cultural Input," in Sperber, Premack, and Premack, pp. 615-44.

59. See *ELT* ; and in film, *Post-Theory* and Noël Carroll, *Theorizing the Moving Image*.

60. Jane Austen's letter to James Edward Austen, December 16, 1816.

61. Letter to Anna Austen, September 9, 1814.

62. George Levine, *Darwin and the Novelists: Patterns of Science in Victorian Fiction* (Cambridge: Harvard University Press, 1988), approaches *Mansfield Park* in a very different way, although the subtitle of his chapter, "Observation Rewarded," could equally be applied to my discussion of the novel. Where I stress evolutionary continuity, he stresses the discontinuities in intellectual history; here, he compares the clear distinctions between right and wrong in Fanny's detached judgments with the clear distinctions and detached observation of "natural theology," as he labels the pre-Darwinian scientific view.

63. Human sexual love "is not a recent invention of the Western leisure classes. People in all cultures experience love and have coined specific words for it": David M. Buss, *The Evolution of Desire: Strategies of Human Mating* (New York: Basic Books, 1994), p. 2. Buss's own surveys cover thirty-seven cultures "from Australia to Zambia" (p. 4) and his findings draw on many more.

64. *The Descent of Man and Selection in Relation to Sex* (London: John Murray, 1871).

65. See Buss, pp. 3, 216 and Matt Ridley, *The Red Queen: Sex and the Evolution of Human Nature* (1993; Harmondsworth: Penguin, 1994), pp. 130-33. The position has altered drastically: see Douglas T. Kenrick, Melanie R. Trost, and Virgil L. Sheets, "Power, Harassment and Trophy Mates: The Feminist Advantages of an Evolutionary Perspective," in *Sex, Power, Conflict: Evolutionary and Feminist Perspectives* (New York: Oxford University Press, 1996), ed. David M. Buss and Neil M. Malamuth, pp. 29-53.

66. "Parental Investment and Sexual Selection," in *Sexual Selection and the Descent of Man* (London: Heinemann, 1972), ed. Bernard Campbell, pp. 136-79, p. 139; see also Douglas T. Kenrick, Edward K. Sadalla, Gary Groth and Melanie R. Trost, "Evolution, Traits, and the Stages of Human Courtship: Qualifying the Parental Investment Model," *Journal of Personality* 58 (1990): 97-116.

67. Buss, p. 22: "The evolution of the female preference for males who offer resources may be the most ancient and pervasive basis for female choice in the animal kingdom."

68. Cited in Ridley, p. 351, n. 21.

69. Buss, pp. 34-35; Pinker and Bloom repeat Symons's "observation that tribal chiefs are often both gifted orators and highly polygynous" (*AM*, p. 483).

70. See Symons, pp. 187-88, Buss, p. 71.

71. Buss reports that "women regard . . . an understanding of the woman's problems" as the "most effective" behavior by which a man can attract a woman (p. 104).

72. See Buss, pp. 102-05, for women's ways of rating men's commitment and concern and men's manipulations in response.

73. P. 170 (see also p. 204); see also Martin Daly and Margo Wilson, "Evolutionary Psychology and Marital Conflict: The Relevance of Stepchildren," in Buss and Malamuth, pp. 9-28, p. 13.

74. As will be the case, Dawkins points out, "if the majority of individuals in a population are the children of experienced . . . mothers--a reasonable assumption in any long-lived species" (*SG*, p. 155). Fanny's moral intelligence more than makes up for her lack of experience.

75. *Mansfield Park*, ed. James Kinsley (Oxford: Oxford University Press, 1990), pp. 428-29 (III.xvii).

76. Nicholas Humphrey suggests that the growth of the human intellect came not from the need to cope with the physical world, with predators or prey, but with the human social world: "The Social Function of Intellect," in *Growing Points in Ethology*, ed. P. P. Bateson and R. A. Hinde, (Cambridge: Cambridge University Press, 1976), pp. 303-18; see also Ridley, pp. 33, 185, 320; de Waal, pp. 24, shows how chimpanzees monitor one another's "moral" performance over time, and p. 33, extends this to humans. For sexual choice and the cognitive arms race, see Buss, pp. 155, 167.

77. See for instance the comic changes in the reactions of the Bertram girls to Henry Crawford, p. 39 (I.v).

78. Buss, p. 20: "Because women in our evolutionary past risked enormous investment as a consequence of having sex, evolution favored women who were highly selective about their mates."

[79](#). Dorrit Cohn, *Transparent Minds: Narrative Modes for Presenting Consciousness in Fiction* (Princeton: Princeton University Press, 1978), p. 108.

[80](#). See Symons, *passim*; Buss, pp. 1, 15.

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